

Amy Flynn, Project #4, 5/4

I. Dataset Information

This dataset includes information about the top 250 movies on IMDB based on their IMDB ratings and can be found here:

<https://www.kaggle.com/datasets/rajugc/imdb-top-250-movies-dataset>. The dataset includes the rank, name, release year, rating, genre, certificate, runtime, tagline, and budget, box-office collection, and casts, writers, and directors. The data was collected in 2021 from IMDB, but it says that it is updated quarterly.

II. My Question

The question that I want to answer is: what makes the best movies on IMDB? I am interested in examining the data and seeing if there is any correlation between specific data categories and their rank on IMDB. Since this is already a dataset of the top 250 movies ranked, I think the best way to do this is to see what characteristics show up the most in this list and try to find a correlation.

III. Code Explanation

```
1 import pandas as pd
2 import plotly.express as px
3
4 rdf = pd.read_csv('./IMDB Top 250 Movies.csv')
5
6 print(rdf)
7 print(rdf.isnull().values.any()) #this returns false
8
```

First, I import pandas and plotly.express, and then pull the file from Kaggle titled 'IMDB Top 250 Movies.csv' and title it rdf for ease. This is a relative file path that points to the next file up from this one in the same folder, so it could be used by someone else easily. Then, I print the cleaned dataset and check to see if there are any missing values with the `.isnull().values.any()` command. This returns False, meaning there are no missing values. As such, that means it does not have to be cleaned so I do not have to rename a cleaned dataset with dropped values because there are no null values to drop.

```

12  #basic statsitics of 250 movie ratings
13  is_ratings=rdf['rating'] #pulls out just rating category values
14  avg_rating=is_ratings.mean()
15  print(avg_rating)
16  min_rating=is_ratings.min()
17  print(min_rating)
18  max_rating=is_ratings.max()
19  print(max_rating)
20

```

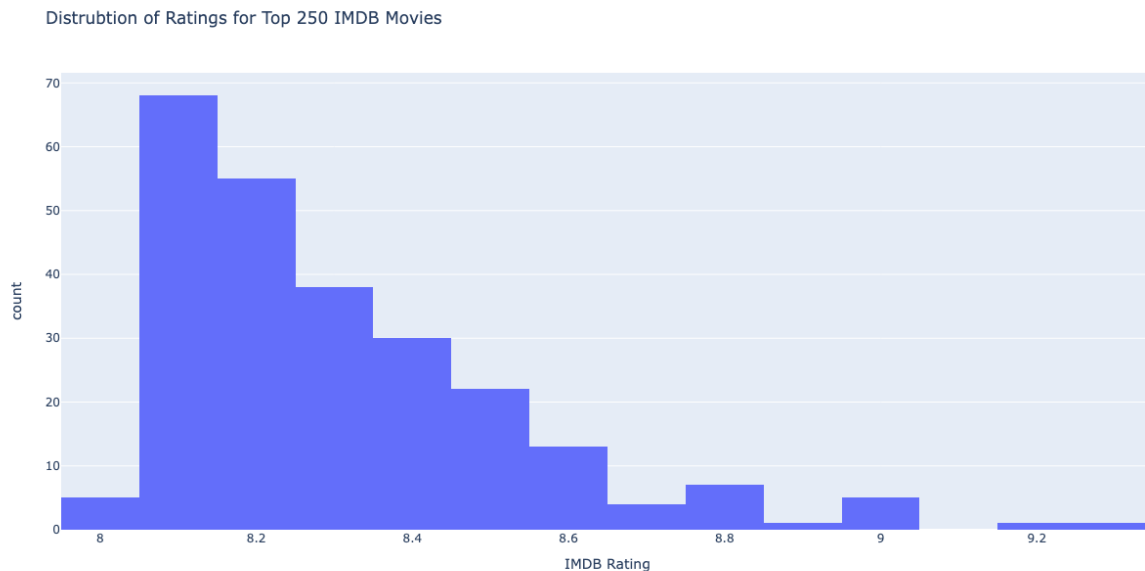
Then, I just printed out some basic statistics of the data set ratings, including the mean, minimum, and maximum values because I wanted to see what distribution of ratings I was working with. That prints out a mean of 8.3072, a minimum of 8.0, and a maximum of 9.3.

```

21  #rating distribution histogram
22  ratingdist=px.histogram(is_ratings, x='rating', nbins=16, title='Distrubtion of Ratings for Top 250 IMDB Movies', labels={"rating": "IMDB Rating", "count": "Frequency"})
23  ratingdist.show()
24

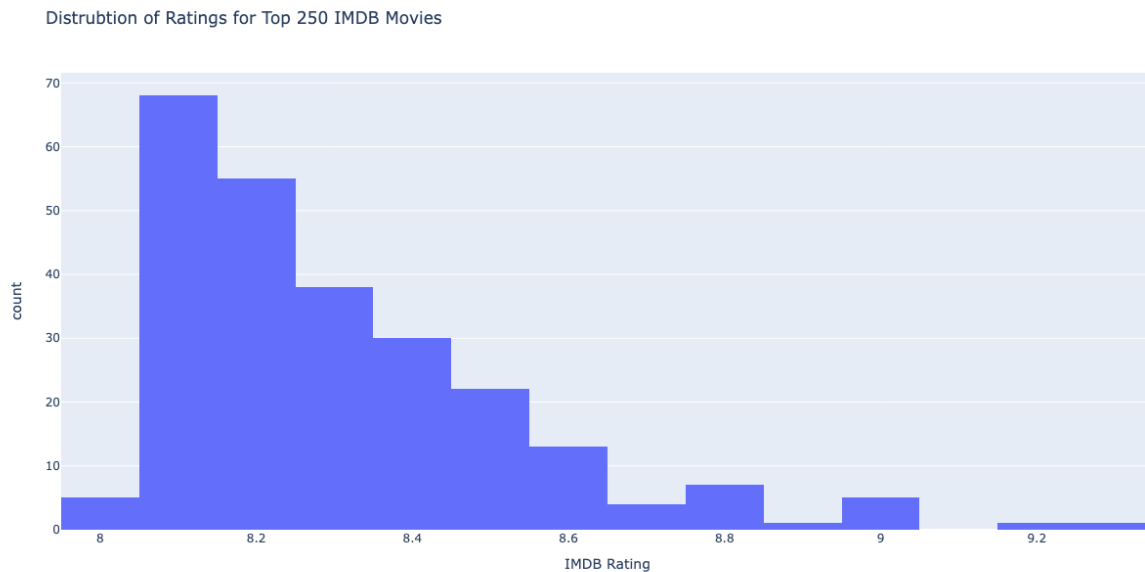
```

In order to see this data more visually, I created a histogram with the data from the rating category with sixteen bins and then fixed the labels and titles. The `.show()` command opens this:



You can see visually that most of the ratings fall between 8.1 and 8.3. There are 161 values included between that range, and out of 250 total values, that is 64% of the total dataset that is in

that distribution. So, now I know that most of the top movies on IMDB are rated between 8.1 and 8.3 out of 10 per critic rating.

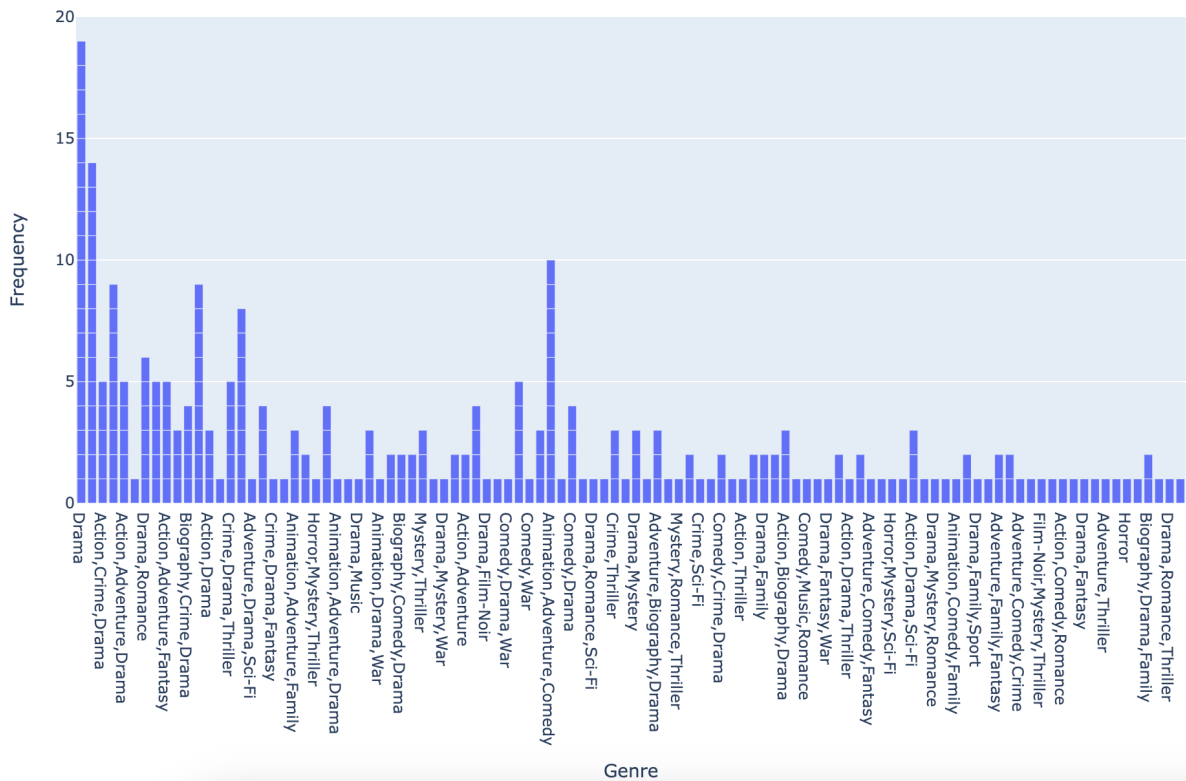


Next, I wanted to know what the most popular genre was for the top 250 movies. So, I made a `is_genre` variable to hold the genre data and used the `mode` function to print out the most often occurring genre value, which prints out as Drama. Then, to see more of this visually, I used `px.bar` to create a bar chart counting the frequency of each genre value that appears. That opens up this graph:

```
25 is_genre=rdf['genre'] #pulls out just genre category values
26 mode_genre = is_genre.mode()
27 print(mode_genre) #prints Drama as most often seen genre in set
28
29 genre_freq = px.bar(is_genre, x='genre', title='Frequency of Genre per Top 250 IMDB Movies', labels={"genre": "Genre", "count": "Frequency"})
30 genre_freq.show() #creates bar chart showing entire layout of genre frequency
```

That opens up this graph, in which you can see that the top five genres for the top 250 movies are Drama, Crime/Drama, Animation/Adventure/Comedy, and tied: Biography/Drama/History, Crime/Drama/Mystery.

Frequency of Genre per Top 250 IMDB Movies



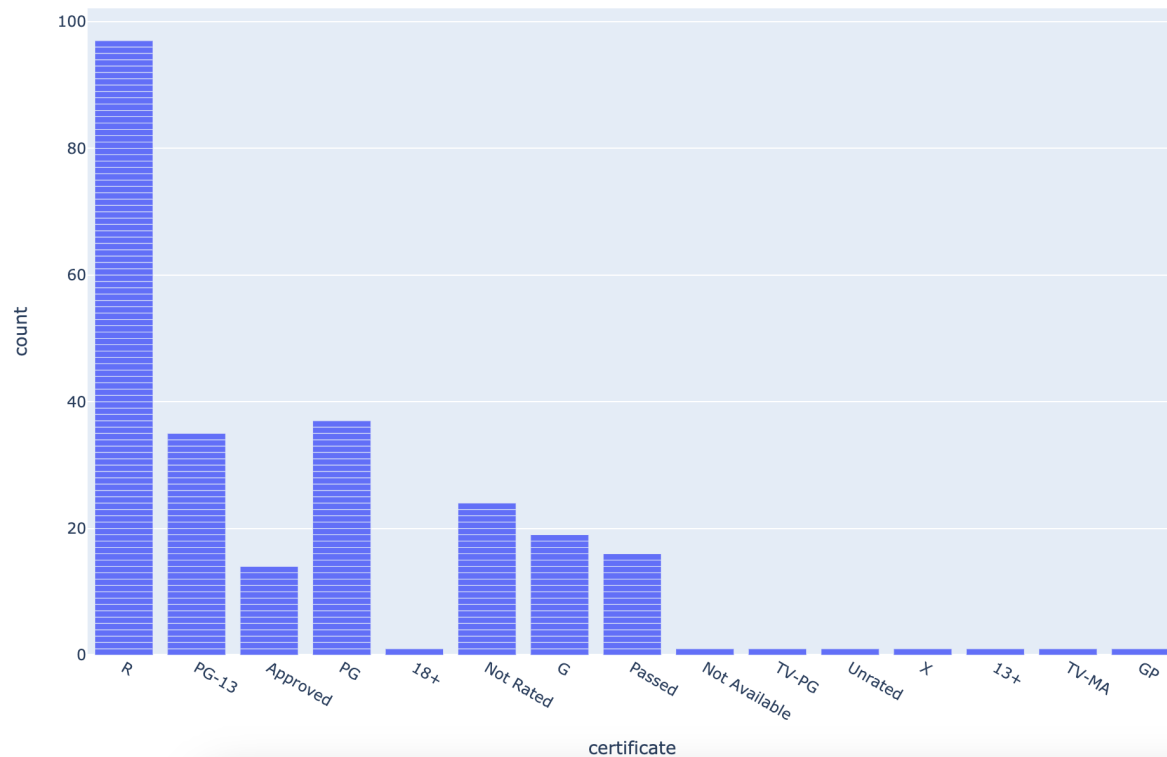
Next, I wanted to know the most popular certificates of each of the top movies, so I created a `certificate_count` variable that counts how many times each certificate shows up in the list. That prints a table in the terminal, but for viewing purposes, I also created a bar chart of the findings. That code and graph are shown here:

```

32 is_certificates=rdf['certificate']
33 certificate_count=is_certificates.value_counts() #counts freq per each certificate value
34 print(certificate_count)
35
36 certificate_freq=px.bar(is_certificates, x='certificate', title='Frequency of Certificate per Top 250 IMDB Movies', labels={"certificate":"Certificate"})
37 certificate_freq.show() #creates bar chart showing freq per certificate
38
39
40 is_budget=rdf["budget"]
41

```

Frequency of Certificate per Top 250 IMDB Movies



Then, I wanted to see if there was any relationship between the budget spent on movies and their rating on IMDB. So, I made a variable called `is_budget` to hold all the budget data and then created a scatterplot. For some reason, my scatterplot was not coming out with the ratings in order, so I did some troubleshooting and research online and found that the `update_layout` function with some specifications could help, and once I included it the ratings went in order! That code and graph is shown below:

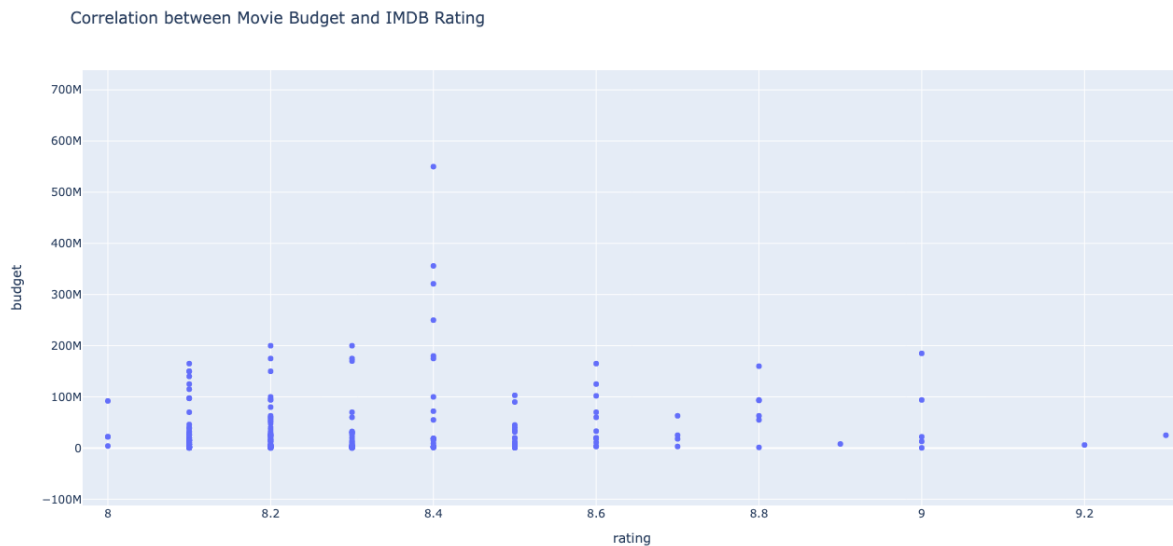
```

40 is_budget=rdf["budget"]
41
42 budget_relationship = px.scatter(rdf, x=is_ratings, y=is_budget, title='Correlation between Movie Budget and IMDB Rating')
43 budget_relationship.update_layout(autotypenumbers='convert types') #my numerical data was not in order for budget - this seemed to help after some research online
44 budget_relationship.show()

```



This original graph had an almost \$2.5 billion outlier, so I zoomed into the graph a bit to see the more accurate distribution. There wasn't much correlation that I could see from the graph, as the highest rated and lowest ratings had such a range of different budgets.



IV. My Findings

My original question was: what makes the best movies on IMDB? I did this by trying to see the most frequent values and distributions of the data for those top 250 movies on the website. Given all the calculations I did, I found that the average ratings of the top movies are between 8.1 and

8.3, the most frequently occurring genre is drama, the most frequently occurring certificate is R, and the budgets are extremely varied between the entire list of movies. As such, I can conclude that the most favorable traits for being ranked in the top 250 movies on IMDB are drama movies rated R and that most of the movies fall between a rating of 8.1 and 8.3 with likely no correlation in budget.